

Supporting Information for : A matter of size: what underpinned the emergence and demise of the largest rodents that ever lived?

Lucas Buffan^{1*}, Myriam Boivin^{2*}, Alicia Álvarez², Fabien L. Condamine¹, Laurent Marivaux¹

Affiliations:

¹ Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France.

² Instituto de Ecorregiones Andinas (INECOA), Universidad Nacional de Jujuy, CONICET, IdGyM, Avenida Bolivia 1661, 4600 San Salvador de Jujuy, Jujuy, Argentina.

³ IANIGLA, CCT-CONICET-Mendoza, Avenida Ruiz Leal s/n, Parque General San Martín, 5500 Mendoza, Argentina.

This file contains:

- Supplementary text
- Supplementary figures
- Supplementary tables

Other Supplementary Materials may be found at <placeholder>.

Codes are available at: <https://github.com/Buffer3369/Chinchilloids.git>

Supplementary text

Some details regarding our occurrence data cleaning

Age ranges of the Miocene occurrences from the Río Santa Cruz (Santacrucian SALMA) were sharpened based on their assigned locality (Arnal et al. 2019). Hence, occurrence from Barrancas Blancas were respectively assigned the 16.3–17.21 and 15.3–16.47 Ma intervals.

Given that *Lagostomus* (*Lagostomopsis*) is a subgenus (Rasia and Candela 2012), we set the **genus** column of each *Lagostomus* (*Lagostomopsis*) occurrence to *Lagostomus*, and switched their **genus_level_status** from **extinct** to **extant**.

All specimens referred to as *Lagostomus minimus*, *L. cavifrons* and *L. heterogenidens* were synonymised with *Lagostomus debilis*, following Ubilla and Rinderknecht (2016).

Uncertain age of the Ituzáingo formation was set to Huayquerian (Late Miocene), *ca.* 5–8 Ma.

Age range of occurrences from Bucher et al. (2021) were set to the one of the original publication: 14.86–12.75 Ma when coming from the “Cruces Infinitos” and “Los Yeguarizos localities”, 12.75–12 Ma when coming from the “La Gloria” and “La Hoyada” localities.

Based on Brandoni et al. (2016), the age range of the occurrences from the Pinturas fauna (*e.g.*, (Kramarz 2002; 2004; 2006)), Burdigalian, were set between 20 and 16.8 Ma.

According to (Esteban et al. 2014; Esteban et al. 2019), the age range of the occurrences belonging to the Andalhualá formation was set to Huayquerian–Chapadmalalan (*c.* 7.14–3.66 Ma).

Perimys perpinguis and *Perimys impactus* were lumped in *Perimys onostus* according to McGrath et al. (2022).

Age range of the occurrence belonging to the Sierra Baguales (Santa Cruz formation, Magallanes, Chile) were set to 19–17.8 Ma according to the isotopic constraints provided in Bostelmann et al. (2013).

Age range of the occurrences reported from the Cerro Boleadoras Formation was set to 16.5–15.1 Ma, according to Vizcaino et al. (2022).

Added two occurrences of *Saremmys ligcura* from the two localities of Bryn Gwyn and Gran Barranca (Chubut, Argentina), according to Busker et al. (2019).

We acknowledge the proposition of synonymisation between *Telicomys gigantissimus* and *Telicomys giganteus* made by Rasia et al. (2025). However, in the light of phylogenetic analyses that are currently in progress, we decided not to account for it, therefore treating *T. gigantissimus* and *T. giganteus* as two distinct taxa.

Supplementary Figures

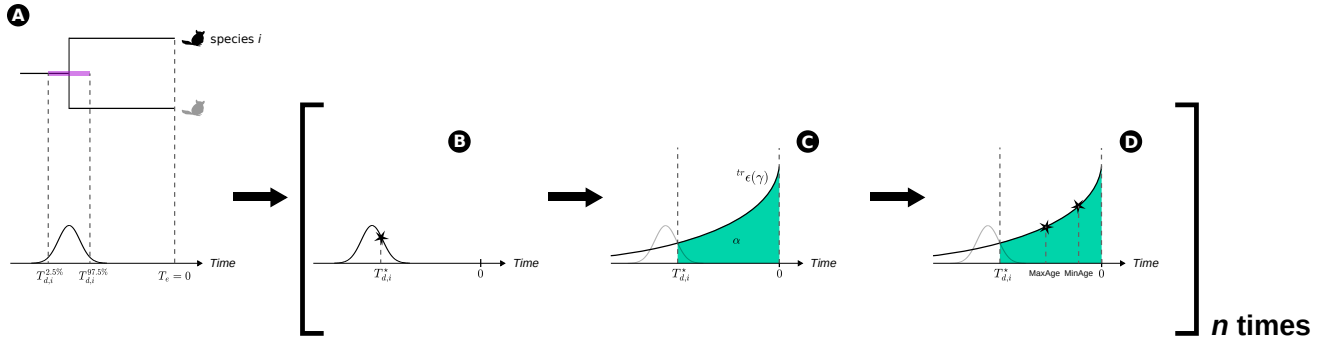


Figure S 1. Visual representation of the occurrence sampling scheme that we implemented for extant species lacking fossil record. We approximated the posterior distribution of the divergence times leading to a given extant species i (denoted $T_{d,i}$) with a gaussian of mean and variance determined by the lower and upper boundaries of the empirical 95% HPD of $T_{d,i}$ (resp. $T_{d,i}^{2.5\%}$ and $T_{d,i}^{97.5\%}$) (A). Then, we randomly drew a value $T_{d,i}^*$ from this distribution, serving as a lower boundary for the species' origination time (B). We then drew simulated occurrence's minimum and maximum ages (resp. 'MinAge' and 'MaxAge') from a truncated exponential distribution (tr_e) of rate γ determined so that a proportion α of the non-truncated distribution lies between 0 and $T_{d,i}^*$ (C–D). Steps B–D were repeated n times, so we finally ended up with n simulated occurrences per extant species lacking fossil record. In practise, we set $n = 10$ and $\alpha = 0.5$, except for *Dinomyx branickii*, for which we set $\alpha = 0.999$. *Chinchilla chinchilla* silhouette is from [Phylopic](#), see Table S2 for additional information.

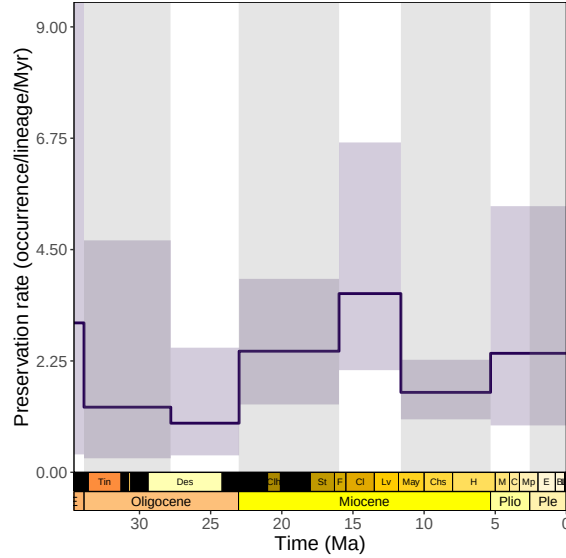


Figure S 2. Estimate of the preservation rate through time in our species-level dataset. The solid line represents the mean posterior estimate and the shaded area the 95% HPD interval.

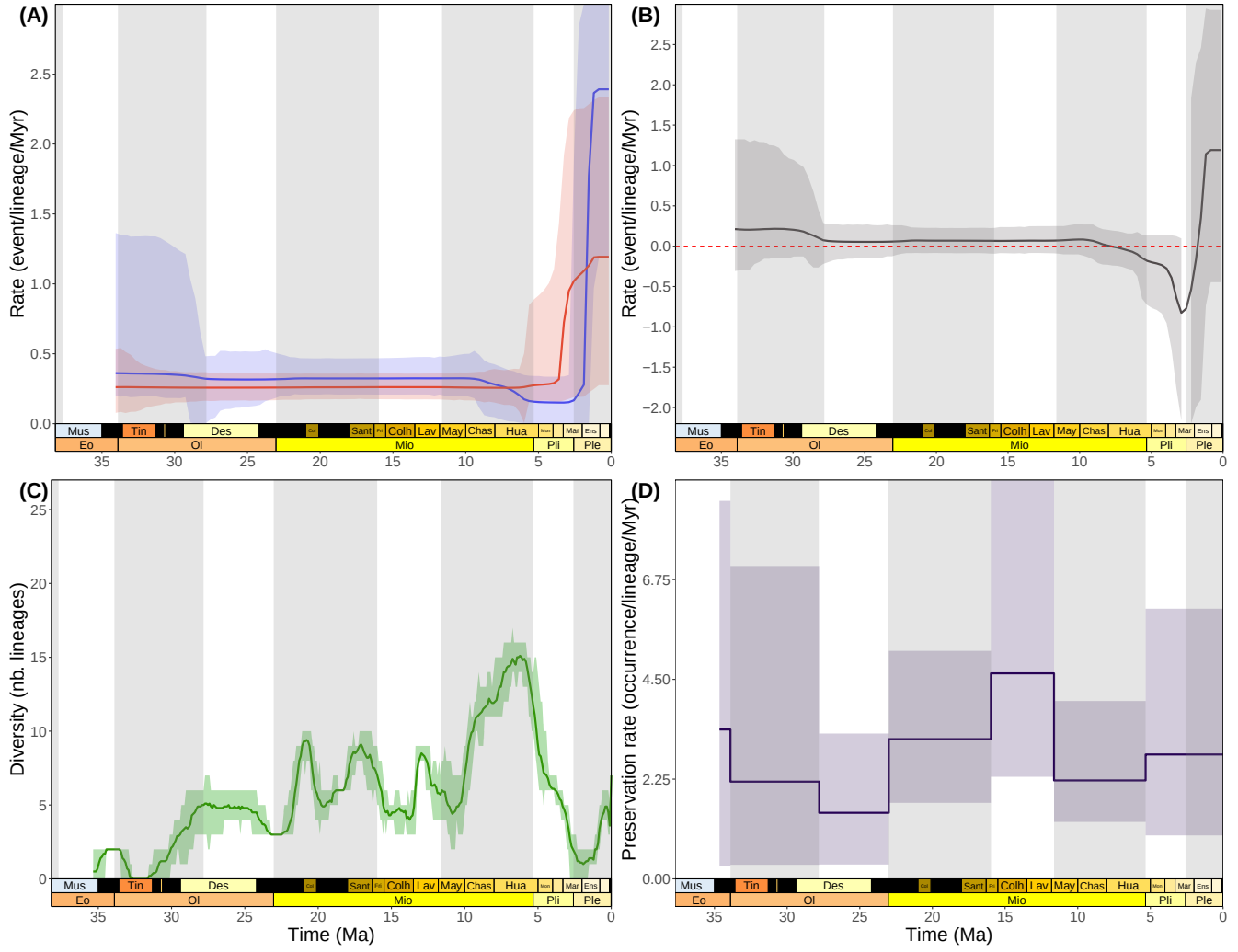


Figure S 3. Speciation and extinction (A), net diversification (= origination - extinction) (B) rates, diversity (C) and preservation rate (D) through time as estimated by the analysis of our species-level occurrence dataset with the rjMCMC model after removing singletons.

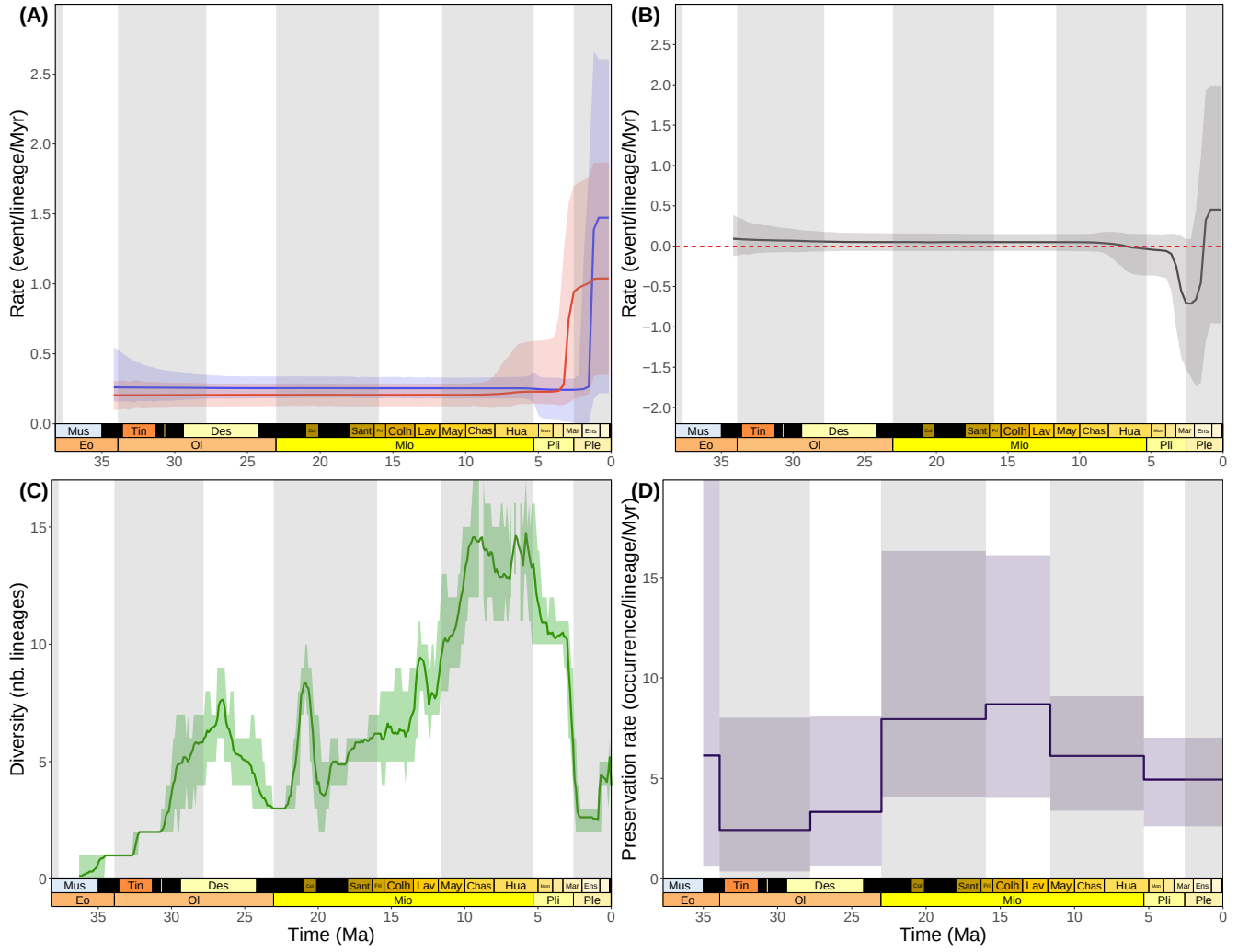


Figure S 4. Origination and extinction (A), net diversification (= origination - extinction) (B) rates, diversity (C) and preservation rate (D) through time as estimated by the analysis of our genus-level occurrence dataset with the rjMCMC model.

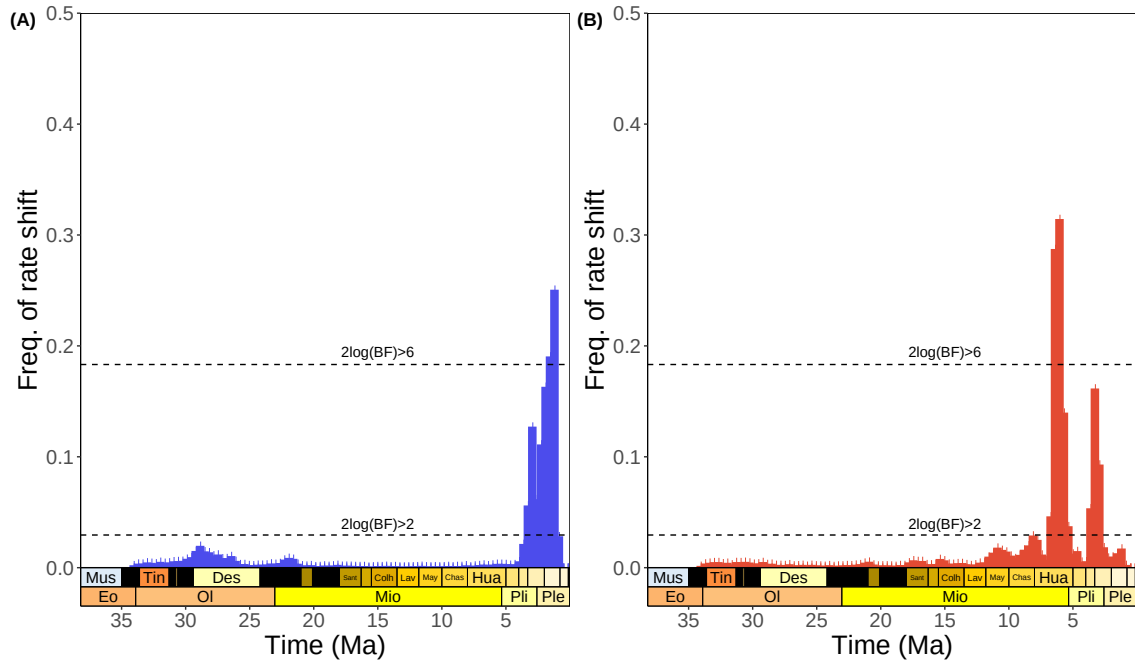


Figure S 5. Frequency of speciation (A) and extinction (B) rate shifts as estimated by the analysis of our species-level chinchilloid occurrence dataset with the rjMCMC model. Dashed horizontal lines indicate $2 * \log(\text{Bayes factor})$ of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

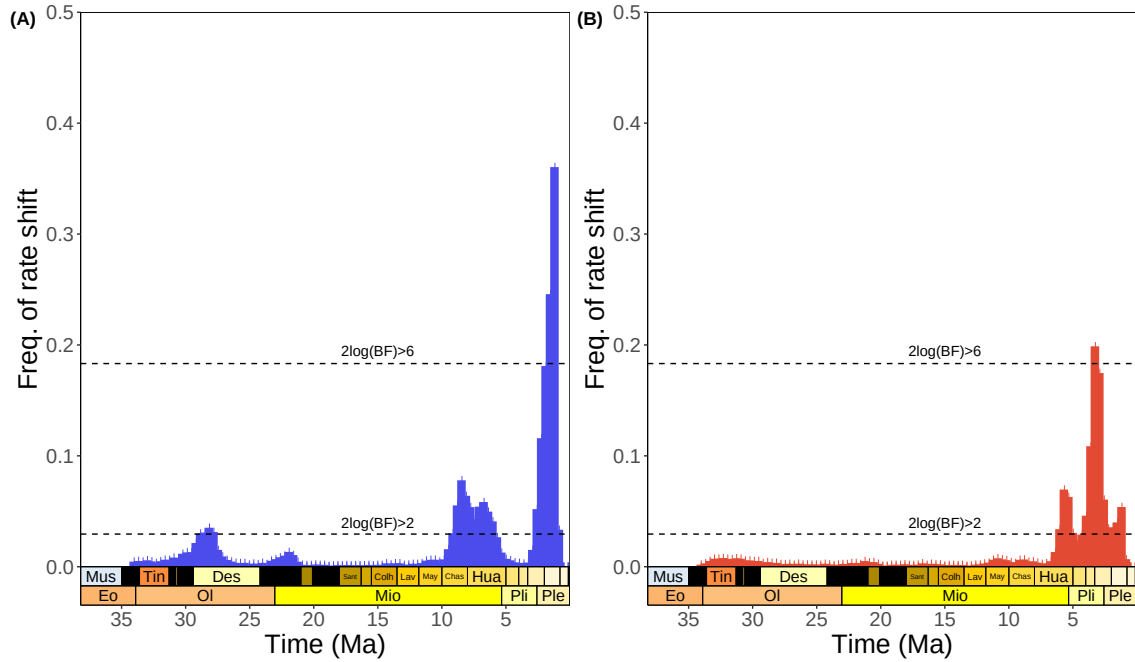


Figure S 6. Frequency of speciation (A) and extinction (B) rate shifts as estimated by the analysis of our species-level chinchilloid occurrence dataset with the rjMCMC model after removing singletons. Dashed horizontal lines indicate $2 * \log(\text{Bayes factor})$ of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

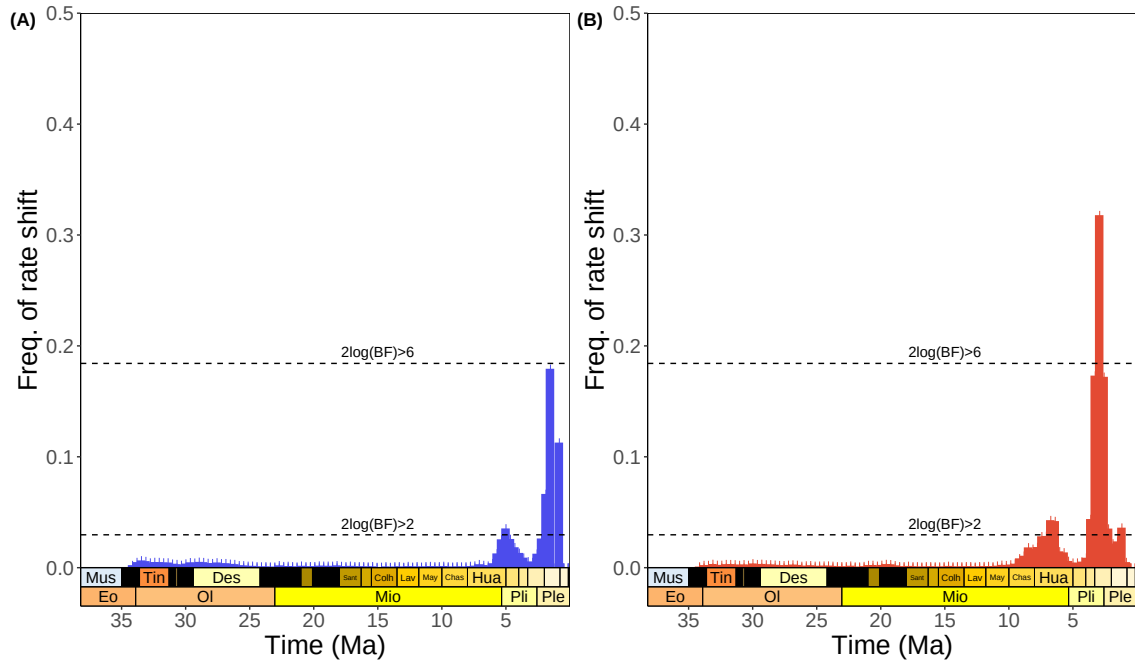


Figure S 7. Frequency of speciation (A) and extinction (B) rate shifts as estimated by the analysis of our genus-level chinchilloid occurrence dataset with the rjMCMC model. Dashed horizontal lines indicate $2 * \log(\text{Bayes factor})$ of 2 and 6, respectively suggesting significant and strongly-supported rate shifts.

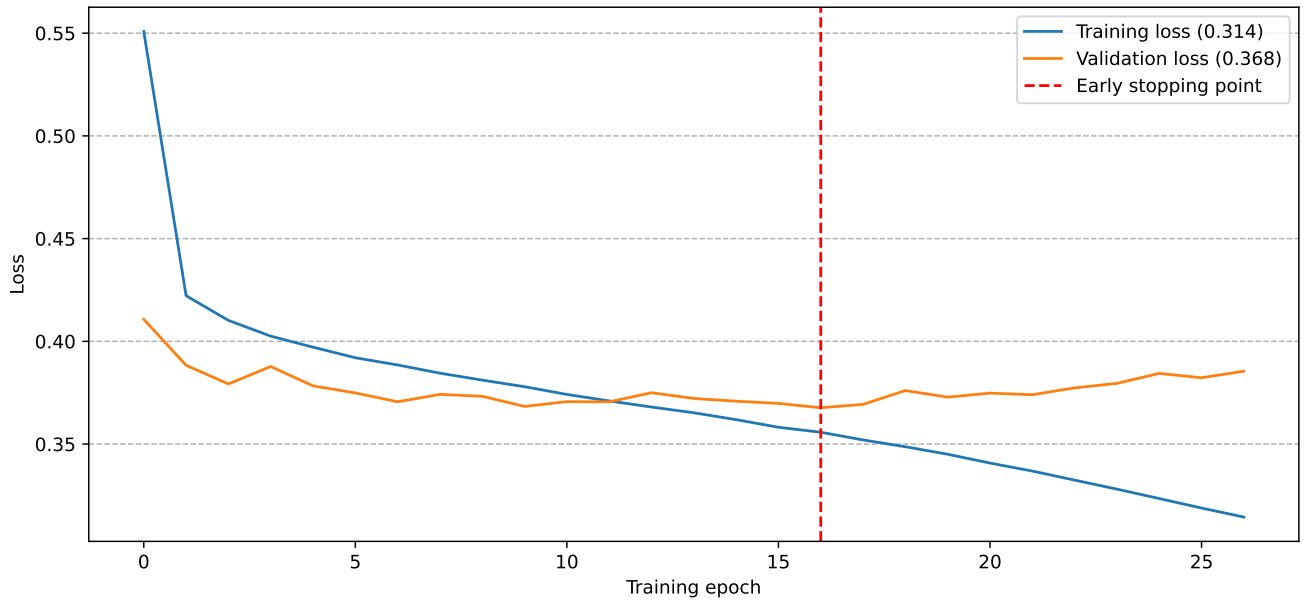


Figure S 8. Training and validation loss across training epochs for a DeepDive network comprising two LSTM units (64 and 32 nodes) connected to two dense layers (64 and 32 nodes) before the output layer. The red vertical line indicates the “early stopping point”, *i.e.*, when the validation loss starts increasing, indicative of overfitting. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.

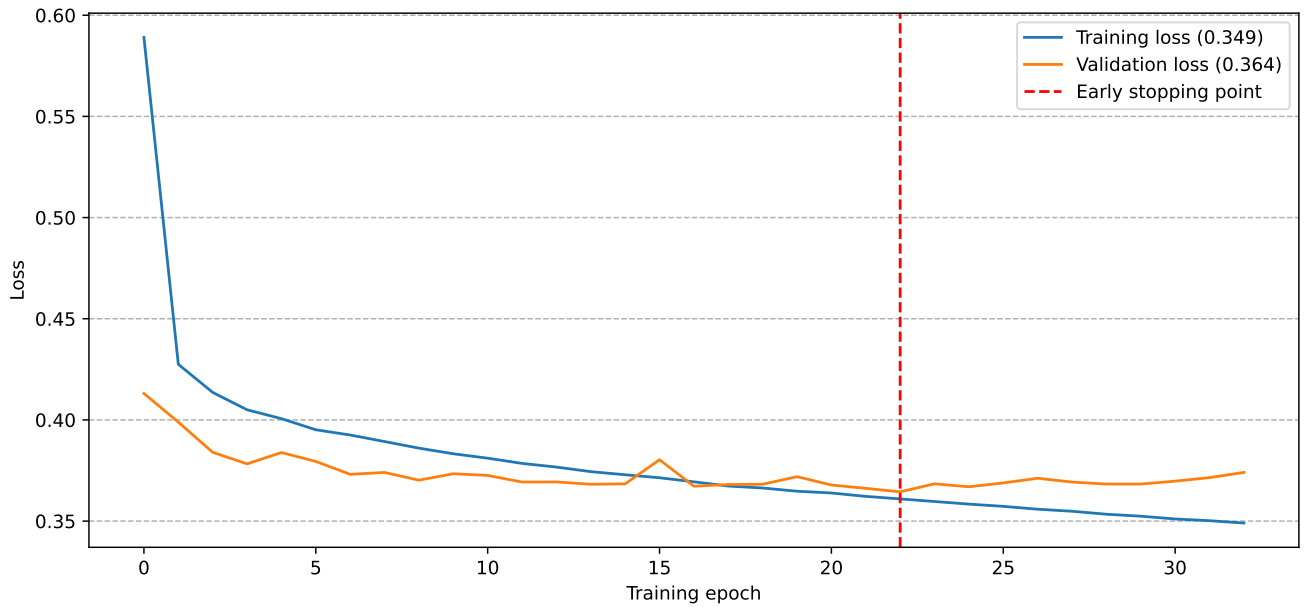


Figure S 9. Training and validation loss across training epochs for a DeepDive network comprising three LSTM units (128, 64 and 32 nodes) connected to two dense layers (64 and 32 nodes) before the output layer. The red vertical line indicates the “early stopping point”, *i.e.*, when the validation loss starts increasing, indicative of overfitting. It determines the optimal value of number of iterations, at which we derive the training and validation loss values depicted between parenthesis.

Supplementary Tables

Table S 1. Performance of the different network architectures tested in our DeepDive analyses. **LSTM Layers** and **Fully-connected** are respectively the number of Long Short Term Memory and Fully-connected units of the network (3: 128, 64 and 32 nodes; 2: 64 and 32 nodes; 1: 32 nodes). **Dropout** represents the Monte-Carlo dropout fraction, that is, the proportion of connections between neurons randomly toggled off between two learning epochs – one epoch being characterised by a complete pass through the entire training dataset –, preventing from overfitting. **Nb. training epochs** is the optimal number of training epochs. **Training loss** and **Validation loss** are the values of the Mean Squared Error (MSE) estimated between the ‘true’ diversity of each simulated dataset (resp. from the training or the validation set), and the predicted diversity made by the network after n rounds of optimisation, where n is the optimal number of training epochs. **Evaluation loss** represent the MSE between each true and its associated predicted diversity trajectory from the test dataset, once the is trained (*i.e.*, after n rounds of optimisation, where n is the optimal number of training epochs). The optimal number of epochs is determined when the **Validation loss** starts increasing (see **?@fig-ddHistPlot**). Architectures resulting in the smallest evaluation loss values, *i.e.*, generating the most accurate predictions, are depicted in bold.

LSTM layers	Fully-connected	Dropout	Nb. training epochs	Training loss	Validation loss	Evaluation loss
3	2	0.05	16	0.314	0.368	0.377
3	1	0.05	14	0.324	0.367	0.38
2	2	0.05	22	0.349	0.364	0.377
2	1	0.05	19	0.354	0.366	0.38
1	1	0.05	41	0.373	0.37	0.381

Table S 2. References of the silhouettes used in Figure 1, from [Phylopic](#).

Silhouette	Author	License
<i>Chinchilla chinchilla</i>	Margot Michaud	CC0 1.0
<i>Lagidium peruanum</i>	Josue Hermoza	CC BY-SA 3.0
<i>Dinomys branickii</i>	Juan C. Cepeda Duque	CC BY 4.0
<i>Josephoartigasia monesi</i>	Nobu Tamura (body), Gustavo Lecuona (head)	CC BY-SA 3.0

Supplementary references

- Arnal, M., M. E. Pérez, and C. Deschamps. 2019. [Revision of the Miocene Caviomorph Rodents From the Río Santa Cruz \(Argentinean Patagonia\)](#). Publicación Electrónica de la Asociación Paleontológica Argentina 19.
- Bostelmann, J. E., J. P. L. Roux, A. Vásquez, N. M. Gutiérrez, J. L. Oyarzún, C. Carreño, T. Torres, R. Otero, A. Llanos, C. M. Fanning, and F. Hervé. 2013. [Burdigalian deposits of the Santa Cruz Formation in the Sierra Baguales, Austral \(Magallanes\) Basin: Age, depositional environment and vertebrate fossils](#). Andean Geology 40:458–489.
- Brandoni, D., L. G. Ruiz, M. F. Tejedor, G. Martin, and J. G. Fleagle. 2016. [Megatherioidea \(Mammalia, Xenarthra, Tardigrada\) from the Pinturas Formation \(Early Miocene\), Santa Cruz Province \(Argentina\) and their chronological implications](#). PalZ 90:619–628.
- Bucher, J., M. E. Pérez, L. R. González Ruiz, L. D'Elía, and A. Bilmes. 2021. [New middle miocene \(langhian - serravallian\) vertebrate localities in northwestern patagonia, argentina: A contribution to high latitude south american land mammal ages sequence](#). Journal of South American Earth Sciences 107:103024.
- Busker, F., M. E. Pérez, and M. T. Dozo. 2019. [A new chinchilloid \(rodentia, hystricognathi\) from the early miocene of the localities of bryn gwyn and gran barranca \(patagonia, argentina\)](#). Comptes Rendus Palevol 18:525–540.
- Esteban, G. I., S. M. Georgieff, N. L. Nasif, L. Ibáñez, and R. A. Bonini. 2019. Marco estratigráfico, paleontología de vertebrados y paleoambientes del plioceno de catamarca, tucumán y santiago del estero, argentina. Pp. 348–366 *in*.
- Esteban, G. I., N. L. Nasif, and S. M. Georgieff. 2014. Cronobioestratigrafía del mioceno tardio-plioceno temprano, puerta de corral quemado y villavil, provincia de catamarca, argentina. Acta geológica lilloan 26:165–192.
- Kramarz, A. 2002. [Roedores chinchilloideos \(Hystricognathi\) de la Formación Pinturas, Mioceno temprano-medio de la provincia de Santa Cruz, Argentina](#). Revista del Museo Argentino de Ciencias Naturales nueva serie 4:167–180.
- Kramarz, A. G. 2006. [Eocardiids \(rodentia, hystricognathi\) from the pinturas formation, late early miocene of patagonia, argentina](#). Journal of Vertebrate Paleontology 26:770–778.
- Kramarz, A. G. 2004. [Octodontoids and erethizontoids \(Rodentia, Hystricognathi\) from the Pinturas Formation, Early - Middle Miocene of Patagonia, Argentina](#). Ameghiniana 41:199–216.
- McGrath, A. J., J. Chick, D. A. Croft, H. E. Dodson, J. J. Flynn, and A. R. Wyss. 2022. [Cavioids, chinchilloids, and erethizontoids \(hystricognathi, rodentia, mammalia\) of the early miocene pampa castillo fauna, chile](#). American Museum Novitates 2022:1–46.
- Rasia, L. L., and A. M. Candela. 2012. [Systematic and biostratigraphic significance of a chinchillid rodent from the pliocene of eastern argentina](#). Acta Palaeontologica Polonica 58:241–254.
- Rasia, L. L., M. de Los Reyes, S. Lorenzini, and A. M. Candela. 2025. [New insights on telicomys giganteus, the last giant dinomyid rodent from the pliocene of central argentina](#). Historical Biology 0:1–18.
- Ubilla, M., and A. Rinderknecht. 2016. [Lagostomus maximus \(desmarest\) \(rodentia, chinchillidae\), the extant plains vizcacha in the late pleistocene of uruguay](#). Alcheringa: An Australasian Journal of Palaeontology 40:354–365.
- Vizcaino, S. F., M. S. Bargo, M. E. Pérez, I. Aramendía, J. I. Cuitiño, E. S. Monsalvo, E. Vlachos, J. I. Noriega, and R. F. Kay. 2022. [Fossil vertebrates of the early-middle Miocene Cerro Boleadoras Formation, northwestern Santa Cruz Province, Patagonia, Argentina](#). Andean Geology 49:382–422.